

Annu Tiwari

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LinkedIn | GitHub | LeetCode | Portfolio

EDUCATION

IILM University, Greater Noida

B.Tech in Computer Science and Engineering

2023 - 2027

CGPA: 8.74/10

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL

Machine Learning & AI: Supervised Learning, Feature Engineering, Data Preprocessing, Model Evaluation, Explainable AI (SHAP), Scikit-learn, NumPy, Pandas, LLM Integration, Prompt Engineering

Web & Backend: Next.js, React.js, Node.js, Express.js, FastAPI, REST APIs, Tailwind CSS, JWT Authentication

Databases & ORM: MongoDB, PostgreSQL, Prisma

Tools & Platforms: Git, GitHub, AWS, VS Code, Postman, Jupyter Notebook, Google Colab, Overleaf, Chrome Extension Development

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

PROJECTS

INORA - Multi-Channel EEG-Based Control

Dec 2025 - Jan 2026

Technologies: AD620, TL072, ADS1115, ESP32, Python, NumPy

- Built a low-cost 3-channel EEG system using AD620, ADS1115, and ESP32 with wireless streaming at **83 Hz/channel**.
- Developed a Python-based EEG processing pipeline with filtering and DRL-based noise reduction, validated on **20 volunteers**, and demonstrated real-time Pong game control.

BrainScore - EEG-Based Depression Detection

Mar 2025 - Nov 2025

Technologies: Python, Pandas, Scikit-learn, LightGBM, SHAP, Google Colab

- Processed EEG recordings from **58 subjects** using **20-channel** signals and extracted **18** statistical, frequency, and entropy-based biomarkers.
- Evaluated six ML models and achieved **95.28%** accuracy with **LightGBM**, using **SHAP** for interpretability.

ApplyPilot - AI-Powered Job Application Platform

Jul 2026 - Present

Technologies: Next.js, TypeScript, React.js, FastAPI, Python, MongoDB, Groq API, JWT

- Engineered a **full-stack AI job application platform** with automated ATS analysis, resume optimization, cover letter generation, job matching, and interview preparation.
- Developed secure **FastAPI REST APIs** with **MongoDB** and built a **browser extension** for extracting job data and connecting it with users' resumes for automated analysis.

EXPERIENCE

AI Engineer Intern

Dec 2025 - Feb 2026

Infosys Springboard

Remote

- Preprocessed cricket match datasets and engineered performance-based features to develop machine learning models for predicting player runs and wickets in upcoming matches.
- Built an interactive **ML dashboard** to visualize player performance, predictions, and match insights.

Machine Learning & Data Science Intern

May 2025 - Jul 2025

Svaarogyam Medical Devices Pvt. Ltd.

Remote

- Developed machine learning workflows for healthcare datasets involving **data preprocessing, feature extraction, exploratory analysis, and predictive modeling**.
- Built end-to-end **machine learning pipelines** for healthcare use cases and contributed to project deliverables within a medical-device development environment.

ACHIEVEMENTS

- Published Patent Application, Indian Patent Office (Application No. 202611071187): Wearable Head-Mounted Device for Continuous Mental State Monitoring and Adaptive Cognitive Regulation
- Research Article accepted in **IEEE Transactions on Medical Robotics and Bionics (T-MRB)**: SHAP-based Study: XAI-Guided Optimisation of Spatial-Temporal Bio-Neuro Marker EEG for Depression
- Research Paper accepted at **IEEE ETNCC 2026**: INORA - A Low-Cost 3-Channel EEG Acquisition System with Driven-Right-Leg Noise Suppression for BCI Prototyping
- Hackathon Winner (**1st Rank**) among 1,004 participants across 118 teams
- Call for Paper Winner**: 1st place at SRM Institute of Science and Technology
- Solved **300+ Data Structures & Algorithms** problems on LeetCode